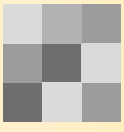
Digital cameras

The digital cameras found in smartphones and tablets, and as stand-alone devices, all share three main features: lenses, which produce an image inside the camera; a light-sensitive chip, or sensor, that captures the image; and a processor that digitizes the image.

How a DSLR works

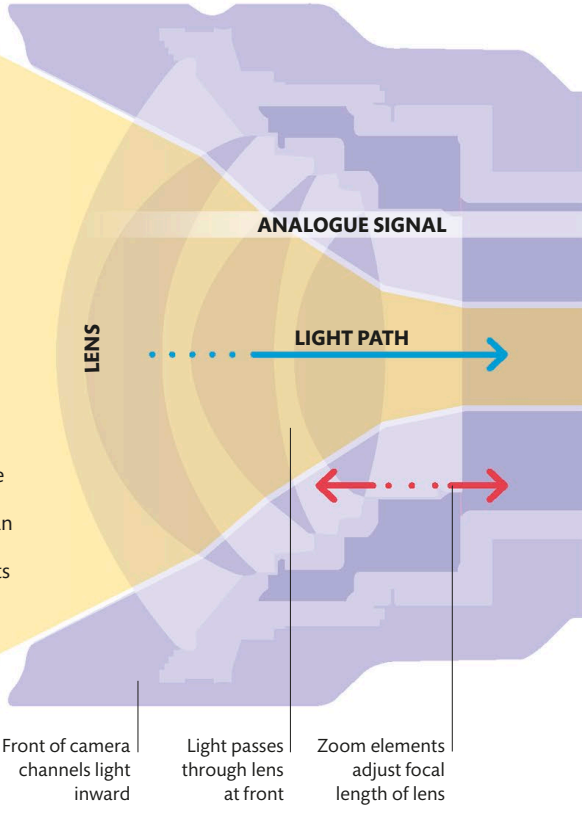
There are two main types of stand-alone digital cameras: compact and DSLR (digital single lens reflex). A compact camera has a main lens and, usually, a separate viewfinder. A DSLR has a mirror that directs light from the main lens up toward an eyepiece lens so that you can see directly through the camera's lens when lining up a shot. The mirror also acts as a shutter, flipping up when the shutter button is released to allow light to fall on the sensor.

THE BIGGEST **DIGITAL IMAGE** IN THE WORLD IS MADE UP OF **365 BILLION PIXELS**, STITCHED TOGETHER FROM **70,000** HIGH-RESOLUTION IMAGES



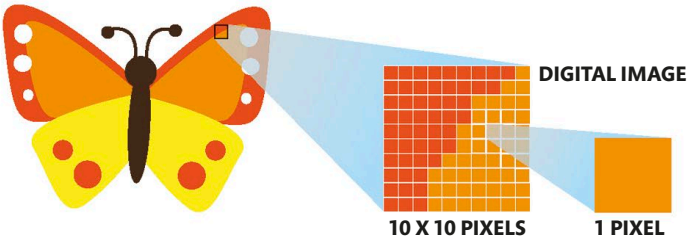
Capturing an image
A camera works a bit like the human eye—with a lens at the front that forms an image at the back. The image falls on an electronic sensor, which has millions of light-sensitive parts arranged in a grid.

1 Focusing light
The lens focuses light to produce an image. It can be moved backward and forward, manually or automatically, to ensure that the subject of the photo is in focus.



PIXELS AND RESOLUTION

A digital image is composed of thousands or millions of dots called picture elements, or pixels. The more pixels, the higher the resolution and the sharper the image. Each pixel has binary numbers associated with it that determine how much red, green, and blue light should be displayed on a screen for that pixel.



WHY ARE PHOTOS TAKEN AT NIGHT OFTEN BLURRY?

In low-light conditions, the shutter needs to stay open longer to collect enough light, so anything that moves during that time will appear blurred.